



Matrix

$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 6 \end{bmatrix}$ 1 row 3 column
 $B = \begin{bmatrix} 3 \\ 4 \\ -5 \end{bmatrix}$ 1 column 3 row

$AB = [2(3) + 5(4) + (6)(-5)]$
 $= [6 + 20 + (-30)]$
 $= [-4]$ 1 row 1 column

$BA = \begin{bmatrix} 6 & 15 & 18 \\ 8 & 20 & 24 \\ -10 & -25 & -30 \end{bmatrix}$ 3 rows 3 column

$3 \times 5 = 15$
 $5 \times 3 = 15$

$1 \times 1 \quad AB = A \times B$
 $3 \times 3 \quad BA = B \cdot A$

$A = \begin{bmatrix} 1 & 4 & -2 \\ 3 & 5 & -6 \end{bmatrix}$ 2 row 3 column
 $B = \begin{bmatrix} 5 & 2 & 8 & -1 \\ 3 & 6 & 4 & 5 \\ -2 & 9 & 7 & -3 \end{bmatrix}$ 3 row 4 column

$AB = \begin{matrix} A & B \\ 2 \times 3 & 3 \times 4 \\ \hline 2 \times 4 \end{matrix}$ same

$BA = \begin{matrix} B & A \\ 3 \times 4 & 2 \times 3 \\ \hline 3 \times 2 \end{matrix}$

BA = doesn't exist

$AB = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 21 & 8 & 10 & 25 \\ 42 & -14 & 2 & 40 \end{bmatrix}$

$1 \times 1 = 1(5) + 4(5) + (-2)(-2) = 21$
 $1 \times 2 = 1(2) + 4(6) + (-2)(9) = 8$
 $1 \times 3 = 1(8) + 4(4) + (-2)(7) = 10$
 $1 \times 4 = 1(-1) + 4(5) + (-2)(3) = 25$
 $2 \times 1 = 3(5) + 5(3) + (-6)(-2) = 42$
 $2 \times 2 = 3(2) + 5(6) + (-6)(9) = -14$
 $2 \times 3 = 3(8) + 5(4) + (-6)(7) = 2$
 $2 \times 4 = 3(-1) + 5(5) + (-6)(-3) = 40$

Combination

A, B, C, D, E, F

$\frac{6 \times 5 \times 4 \times 3}{1 \times 2 \times 3 \times 4} = \frac{6!}{(6-4)!} = \frac{6 \times 5 \times 4 \times 3 \times 2 \times 1}{2 \times 1} = 360$

of ways to arrange k things in k spots

$n P_k = \frac{n!}{(n-k)!}$

- ① A B C D } order matter
- ② D A B C } two difference form

$\frac{k}{1} \frac{k-1}{2} \frac{k-2}{3} \dots \frac{1}{k} = k!$

$n C_k$ ↓ order doesn't matter 2 form above is same form

$n C_k = \frac{n!}{(n-k)! k!} = \frac{n!}{k!(n-k)!}$

${}^6 C_4 = \binom{6}{4} = \frac{6!}{4!(6-4)!} = \frac{6!}{4!(2!)} = \frac{6 \times 5 \times 4 \times 3 \times 2 \times 1}{4 \times 3 \times 2 \times 1 \times 2 \times 1} = \frac{30}{2} = 15$

${}^{10} C_3 = \binom{10}{3} = \frac{10!}{3!(10-3)!} = \frac{10!}{3!(7!)} = \frac{10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{3 \times 2 \times 1 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1} = \frac{720}{6} = 120$

Practice Questions

Level 1 understanding the concept

- if $n=6, k=2$, is this a valid column? = 1, 0, 1, 0, 0, 0 ✓
- if $n=6, k=2$, is this a valid column? = 1, 1, 1, 0, 0, 0 X
- if $n=6, k=2$, is this a valid column? = 0, 0, 0, 0, 0, 0 X

${}^6 C_2 \binom{6}{2} = \frac{6!}{2!(6-2)!} = \frac{6 \times 5}{2} = 15$

Modulus study

$a \equiv b \pmod{n}$ if and only if $n \mid a-b$ (n divides $a-b$)

a, b and n are integers; $n > 1$

Ex: $5 \equiv 2 \pmod{3}$

$3 \mid 5-2$

$a-b = nk$

$4 \equiv -1 \pmod{5}$

$5 \mid 4 - (-1)$

Ex. Find all possible values of n if $11 \equiv 5 \pmod{n}$

$n \mid 11-5$

$11 \equiv 5 \pmod{6}$



• Modular Arithmetic deals with remainders

• Divide a by n and let the quotient be q and the remainder r , then we have

$a = nq + r$ which implies $a \equiv r \pmod{n}$

Examples

13 divide by 5

$$\begin{array}{r} 2 \\ 5 \overline{) 13} \\ \underline{10} \\ 3 \end{array}$$

quotient is 2 (q)

Remainder is 3 (r)

$13 \equiv 3 \pmod{5}$

$2 \equiv -3 \pmod{5}$

$5 \mid 2 - (-3)$

$-8 \equiv 7 \pmod{5}$

$5 \mid -8 - 7$

$5 \mid -15$

$-3 \equiv -8 \pmod{5}$

$5 \mid -8 - (-3)$

$5 \mid -5$

We can use addition, subtraction, and multiplication but division is problematic

Ex: Solve $x^2 + 3 \equiv 0 \pmod{7}$

$7 \equiv 0 \pmod{7}$

$x^2 \equiv -3 \pmod{7}$

$n \equiv 0 \pmod{n}$

$x^2 \equiv 4 \pmod{7}$

$\left\{ \begin{array}{l} x \equiv 2 \pmod{7} \\ x \equiv -2 \pmod{7} \end{array} \right.$

$nk \equiv 0 \pmod{n}$

$\{0, 1, 2, 3, 4, 5, 6\}$

Ex: $10 \equiv 4 \pmod{6}$

$6 \mid 10-4$

$10 \equiv 4 \pmod{6}$

$\downarrow$
 $5 \equiv 2 \pmod{6}$

Ex. $x^2 \equiv 2 \pmod{4} \rightarrow \{0, 1, 2, 3\}$
 $\downarrow \downarrow \downarrow \downarrow$
 $\{0, 1, 0, 1\} \leftarrow x^2$

No solution

What about negative exponents

$3x \equiv 1 \pmod{5}$

$\rightarrow 3x \equiv 6 \pmod{5}$

$3^{-1} \cdot 3x = 3^{-1} \cdot 1$

$x \equiv 2 \pmod{5}$

$x \equiv 3^{-1}$

$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, B = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

Transpose of matrix $\equiv$

$$\begin{matrix} n \\ m \end{matrix} \begin{matrix} A \\ \end{matrix} = \begin{matrix} \begin{matrix} a_{1,1} & a_{1,2} & \dots & a_{1,n} \\ a_{2,1} & a_{2,2} & \dots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m,1} & a_{m,2} & \dots & a_{m,n} \end{matrix} \\ m \end{matrix}$$

$$\begin{matrix} n \\ n \end{matrix} \begin{matrix} A^T \\ \end{matrix} = \begin{matrix} \begin{matrix} a_{1,1} & a_{2,1} & \dots & a_{m,1} \\ a_{1,2} & a_{2,2} & \dots & a_{m,2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1,n} & a_{2,n} & \dots & a_{m,n} \end{matrix} \\ \text{transpose of } A \end{matrix}$$

$\equiv$ Full-Rank Matrix $\equiv$

(for binary matrix)

↑ the element of the matrix is $\in \{0,1\}$

• Permutation matrix $\begin{matrix} n \\ n \end{matrix}$

a square binary matrix s.t.
there is only one non-zero element
in each row and each column

e.g. $\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \rightarrow$ Permutation matrix

$\begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \rightarrow$ NOT permutation matrix

$\begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix} \rightarrow$ NOT

$(A \times B^T) = B^T \times A^T$ check

full-rank matrix

a binary matrix $A \in \{0,1\}^{m \times n}$ is a full-rank matrix

if: There exists a permutation matrix P , a square matrix L and a matrix U
s.t.

$A = PLU$

$$\begin{matrix} m \\ m \end{matrix} \begin{matrix} P \\ \end{matrix} \begin{matrix} m \\ n \end{matrix} \begin{matrix} A \\ \end{matrix} = \begin{matrix} m \\ n \end{matrix} \begin{matrix} L \\ \end{matrix} \begin{matrix} m \\ m \end{matrix} \begin{matrix} U \\ \end{matrix}$$

$$\begin{matrix} \begin{bmatrix} 1 & 0 & \dots & 0 \\ & 1 & 0 & \dots & 0 \\ & & \ddots & & \\ & & & 0 & 1 \\ & & & & 1 \end{bmatrix} & \begin{bmatrix} 1 & * & \dots & * \\ 0 & \dots & * & \\ \vdots & & & 1 \\ 0 & \dots & 0 & 1 \\ 0 & \dots & 0 & 0 \end{bmatrix} \end{matrix} \Bigg\} n$$

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

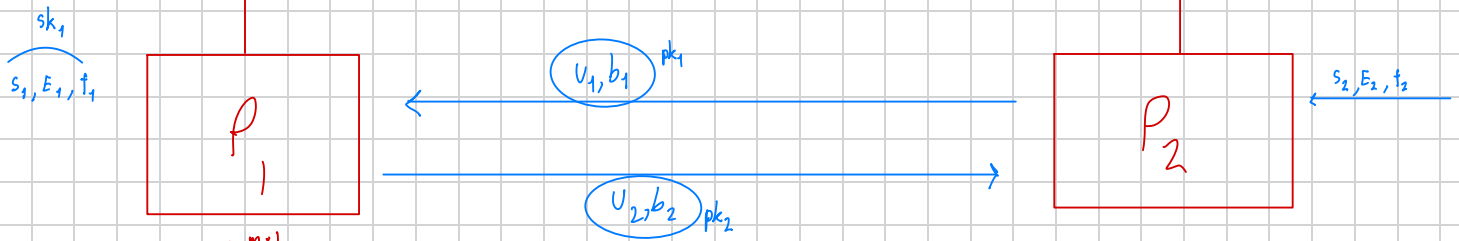
$$= \begin{matrix} L & U \end{matrix} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \Bigg\} n=3$$

rank(A) = 3

if row(A) = n $\Rightarrow$ A is full-rank

$$m = 2n$$

$A: f_{ij}$ rank random A is a random matrix with m rows n columns elements in $\{0,1\}$
binary matrix $\in \{0,1\}$



$$① f_1 \leftarrow \{0,1\}^{m \times 1}$$

$$f_1 = \begin{bmatrix} 1 \\ \vdots \\ m \end{bmatrix}$$

f has k non-zero elements $\left. \begin{array}{l} \text{random} \\ n-k \text{ zero elements} \end{array} \right\}$

hopefully
($d_2 \neq d_1$)

create f_2, s_2, E_2 with similar procedure as f_1, s_1, E_1

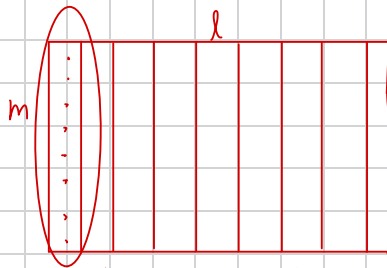
$$\begin{aligned} U_2 &= A S_2 + E_2 \text{ mod } 2 \\ k &= A^T f \text{ mod } 2 \\ k_2^T &= U_1^T f_2 + S_2^T b_1 \text{ mod } 2 \\ k_2 &= \text{maj}(k_2) \end{aligned}$$

$$② S_1 \leftarrow \{0,1\}^{n \times l} \text{ (random)}$$

$$③ E_1 \leftarrow \{0,1\}^{m \times l}$$

each column has

$\left(\begin{array}{l} k \text{ non-zero elements} \\ n-k \text{ zero elements} \end{array} \right) \text{ random}$
Put 0 into



$\left(\begin{array}{l} k \text{ non-zero elements} \\ n-k \text{ zero elements} \end{array} \right) \text{ random for each row}$

$$④ U_1 = A S_1 + E_1 \text{ mod } 2$$

$$b_1 = A^T f_1 \text{ mod } 2$$

$$⑤ k_1 = U_2^T f_1 + S_1^T b_2 \text{ mod } 2$$

$$⑥ \tilde{K}_1 = \text{maj}(k_1) =$$

$$\text{maj}(k_1) = \begin{cases} 1 & \text{if the number of non-zero elements on } k_1 \geq \frac{l}{2} \\ 0 & \text{otherwise} \end{cases}$$

example $l = 8$

$$k_1 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} \quad \text{maj}(k_1) = 0$$

(because the number of non-zero elements $\leq \frac{l}{2}$)

$$k_1 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 1 \\ 1 \\ 1 \\ 0 \end{bmatrix} \quad \text{maj}(k_1) = 0$$

(because the number of non-zero elements $\leq \frac{l}{2}$)

Experiment 1

Execute the key-exchange 10,000 times

check how many times $\tilde{k}_1 = \tilde{k}_2$

$$\text{maj}(k_1) = \text{maj}(k_2)$$

Task

repeat the experiment 1

10,000 times, and find the average

number of times that $\text{maj}(k_1) = \text{maj}(k_2)$

$$* n = 2^8 = 256$$

$$m = 2 \cdot n = 2^9 = 512$$

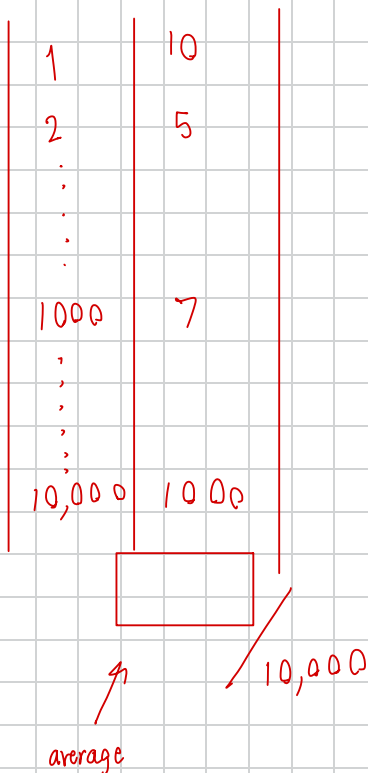
$$k = \frac{1}{\sqrt{n}} \cdot m = 2^5 = 32$$

$$l = 1000$$

$$l = 1024 \text{ (new)}$$

$$k = 16 \text{ (new)}$$

parameter we gonna use



exp1.csv ... exp10000.csv

run		$\text{maj}(k_1^{(i)})$	$\text{maj}(k_2^{(i)})$
1	$k_1^{(1)}, k_2^{(1)}$	$\text{maj}(k_1^{(1)})$	$\text{maj}(k_2^{(1)})$
2	$k_1^{(2)}, k_2^{(2)}$	$\text{maj}(k_1^{(2)})$	$\text{maj}(k_2^{(2)})$
3	$k_1^{(3)}, k_2^{(3)}$	$\text{maj}(k_1^{(3)})$	$\text{maj}(k_2^{(3)})$
...	...		
10	$k_1^{(10)}, k_2^{(10)}$		
...	...		
10,000	$k_1^{(10,000)}, k_2^{(10,000)}$		

Overall.csv

exp		
1	$\text{maj}(\text{maj}(k_1^{(1)}), \dots, \text{maj}(k_1^{(10,000)}))$ in file exp1.csv	$\text{maj}(\text{maj}(k_2^{(1)}), \dots, \text{maj}(k_2^{(10,000)}))$ in file exp1.csv
...		
...		
...		
10,000	exp. 10000.csv run/experiment	

Transpose

$$A = \begin{bmatrix} 1 & -2 \\ 3 & 0 \\ 7 & 5 \end{bmatrix} \begin{matrix} 1 \\ 2 \\ 3 \end{matrix}$$

3×2

$$A^T = \begin{bmatrix} 1 & 3 & 7 \\ -2 & 0 & 5 \end{bmatrix} \begin{matrix} 2 \\ 3 \end{matrix}$$

Transpose of A

rows $\Rightarrow$ columns
columns $\Rightarrow$ row

$$A^T = \begin{bmatrix} 1 & 3 & 7 \\ -2 & 0 & 5 \end{bmatrix}$$

Properties

Proof of properties of ④

$$(AB)_{ij} = \sum_{k=1}^n a_{ik} b_{kj}$$

$$\text{Check: } (AB)^T_{ji}$$

$$= (AB)_{ji}$$

$$= \sum_{k=1}^n a_{jk} b_{ki}$$

$$= \sum_{k=1}^n b_{ki} a_{jk}$$

$$= \sum_{k=1}^n (b^T)_{ik} (A^T)_{kj}$$

$$= (b^T A^T)_{ij}$$

$$\textcircled{1} (A^T)^T = A$$

$$\textcircled{2} (A+B)^T = A^T + B^T$$

$$\textcircled{3} (kA)^T = k(A^T)$$

$$\textcircled{4} (AB)^T = B^T A^T$$

$$\textcircled{5} \det(A^T) = \det A$$

$$\textcircled{6} (A^{-1})^T = (A^T)^{-1}$$

Recall identity matrix

- Is square matrix

- Consists of only 1's and 0's

- The principal diagonal are 1's and all other elements are 0's.

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \times \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

Matrix A Matrix I Matrix A

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

Matrix I Matrix A

$$A \times I = I \times A = A$$

Permutation Summary

Column permutation - switch columns

$$B = A \times P$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \times \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 3 & 2 & 1 \\ 6 & 5 & 4 \\ 9 & 8 & 7 \end{bmatrix}$$

Matrix A Matrix P Matrix B

Row permutation - switch rows

$$B = P \times A$$

$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \times \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} = \begin{bmatrix} 7 & 8 & 9 \\ 4 & 5 & 6 \\ 1 & 2 & 3 \end{bmatrix}$$

LU decomposition

$$A = \begin{pmatrix} 2 & 4 & 3 & 5 \\ -4 & -7 & -5 & -9 \\ 6 & 8 & 2 & 9 \\ 4 & 9 & -2 & 14 \end{pmatrix} \quad (4 \times 4 \text{ matrix})$$

$$= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 4 & 3 & 5 \\ -4 & -7 & -5 & -9 \\ 6 & 8 & 2 & 9 \\ 4 & 9 & -2 & 14 \end{pmatrix} \begin{matrix} \\ \text{II} - (-2) \cdot \text{I} \\ \text{III} - 3 \cdot \text{I} \\ \text{IV} - 2 \cdot \text{I} \end{matrix}$$

$$= \begin{pmatrix} 1 & & & \\ -2 & 1 & & \\ 3 & & 1 & \\ 2 & & & 1 \end{pmatrix} \begin{pmatrix} 2 & 4 & 3 & 5 \\ 0 & 1 & 1 & 2 \\ 0 & -4 & -7 & -6 \\ 0 & 1 & -8 & 4 \end{pmatrix} \begin{matrix} \\ \\ \text{III} - (-4) \cdot \text{II} \\ \text{IV} - (1) \cdot \text{II} \end{matrix}$$

$$= \begin{pmatrix} 1 & & & \\ -2 & 1 & & \\ 3 & -4 & 1 & \\ 2 & 1 & & 1 \end{pmatrix} \begin{pmatrix} 2 & 4 & 3 & 5 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & -3 & 2 \\ 0 & 0 & -9 & 2 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & & & \\ -2 & 1 & & \\ 3 & -4 & 1 & \\ 2 & 1 & 3 & 1 \end{pmatrix} \begin{pmatrix} 2 & 4 & 3 & 5 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & -3 & 2 \\ 0 & 0 & 0 & -4 \end{pmatrix} \left. \begin{matrix} \\ \\ \\ \text{IV} - (-3) \cdot \text{III} \end{matrix} \right\} U = A$$

L

PLU decomposition
 lower triangular
 permutation matrix
 row echelon

$$A = \begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 0 & 0 & 0 & 1 \\ 2 & 1 & 2 & 1 & 5 \\ 1 & 2 & 4 & 3 & 6 \end{pmatrix} = P_{1 \leftrightarrow 2}^2 \begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ \boxed{1} & 0 & 0 & 0 & 1 \\ 2 & 1 & 2 & 1 & 5 \\ 1 & 2 & 4 & 3 & 6 \end{pmatrix}$$

$$= P_{1 \leftrightarrow 2} \begin{pmatrix} \boxed{1} & 0 & 0 & 0 & 1 \\ 0 & 1 & 2 & 1 & 2 \\ 2 & 1 & 2 & 1 & 5 \\ 1 & 2 & 4 & 3 & 6 \end{pmatrix}$$

$$P_{1 \leftrightarrow 2} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \boxed{1} & 0 & 0 & 0 & 1 \\ \boxed{0} & 1 & 2 & 1 & 2 \\ \boxed{2} & 1 & 2 & 1 & 5 \\ 1 & 2 & 4 & 3 & 6 \end{pmatrix} \begin{matrix} \\ \text{II} - 0\text{I} \\ \text{III} - 2\text{I} \\ \text{IV} - 1\text{I} \end{matrix}$$

$$P_{1 \leftrightarrow 2} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 2 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \boxed{1} & 0 & 0 & 0 & 1 \\ 0 & \boxed{1} & 2 & 1 & 2 \\ 0 & \boxed{1} & 2 & 1 & 3 \\ 0 & \boxed{2} & 4 & 3 & 5 \end{pmatrix} \begin{matrix} \\ \\ \text{III} - 1\text{II} \\ \text{IV} - 2\text{II} \end{matrix}$$

$$P_{1 \leftrightarrow 2} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 2 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{pmatrix} P_{3 \leftrightarrow 4}^2 \begin{pmatrix} \boxed{1} & 0 & 0 & 0 & 1 \\ 0 & \boxed{1} & 2 & 1 & 2 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & \boxed{1} & 1 \end{pmatrix}$$

$$P_{1 \leftrightarrow 2} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 2 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{pmatrix} P_{3 \leftrightarrow 4}^2 \begin{pmatrix} \boxed{1} & 0 & 0 & 0 & 1 \\ 0 & \boxed{1} & 2 & 1 & 2 \\ 0 & 0 & 0 & \boxed{1} & 1 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

$$P_{1 \leftrightarrow 2} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 \\ 2 & 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} \boxed{1} & 0 & 0 & 0 & 1 \\ 0 & \boxed{1} & 2 & 1 & 2 \\ 0 & 0 & 0 & \boxed{1} & 1 \\ 0 & 0 & 0 & 0 & \boxed{1} \end{pmatrix}$$

P L U

Rank of matrix

The rank of matrix is the number of linearly independent rows.

Or equivalently, the number of linearly independent columns. Only a zero matrix has rank zero.

To be in row echelon form, a matrix must have the following properties:

1. If a row doesn't consist entirely of zeros, then the first non-zero number in the row is a 1.
2. All zero rows are at the bottom of the matrix.
3. In any two successive rows that do not consist entirely of zeros, the leading 1 in the lower row occurs farther to the right than the leading 1 in the higher row.

Rank matrix

Step ① Verify $U1 = A \cdot S1 + E1 \text{ mod } 2$

$$A: \begin{bmatrix} 0 & 1 \\ 1 & 1 \\ 0 & 0 \\ 1 & 0 \end{bmatrix} \quad S_1: \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A \cdot S1 \text{ row1: } \begin{bmatrix} 0 & 1 \end{bmatrix} \times \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0, 1 \end{bmatrix}$$

$$+ E1 \text{ row1: } \begin{bmatrix} 0, 1 \end{bmatrix} = \begin{bmatrix} 0+0, 1+1 \end{bmatrix} = \begin{bmatrix} 0, 2 \end{bmatrix} = \begin{bmatrix} 0, 0 \end{bmatrix}$$

$$A \cdot S1 \text{ row2: } \begin{bmatrix} 1, 1 \end{bmatrix} \times \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0, 1 \end{bmatrix}$$

$$+ E1 \text{ row2: } \begin{bmatrix} 1, 0 \end{bmatrix} = \begin{bmatrix} 1, 0 \end{bmatrix} + \begin{bmatrix} 0, 1 \end{bmatrix} = \begin{bmatrix} 1, 1 \end{bmatrix}$$

$$A \cdot S1 \text{ row3: } \begin{bmatrix} 0, 0 \end{bmatrix} \times \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0, 0 \end{bmatrix}$$

$$+ E1 \text{ row3: } \begin{bmatrix} 0, 0 \end{bmatrix} = \begin{bmatrix} 0, 0 \end{bmatrix} + \begin{bmatrix} 0, 0 \end{bmatrix} = \begin{bmatrix} 0, 0 \end{bmatrix}$$

$$A \cdot S1 \text{ row4: } \begin{bmatrix} 1, 0 \end{bmatrix} \times \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0, 0 \end{bmatrix}$$

$$+ E1 \text{ row4: } \begin{bmatrix} 0, 0 \end{bmatrix} = \begin{bmatrix} 0, 0 \end{bmatrix} + \begin{bmatrix} 0, 0 \end{bmatrix} = \begin{bmatrix} 0, 0 \end{bmatrix}$$

Step 3 Verify $U2 = A \cdot S2 + E2 \text{ mod } 2$

$$A: \begin{bmatrix} 0 & 1 \\ 1 & 1 \\ 0 & 0 \\ 1 & 0 \end{bmatrix}$$

$$S_2: \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$$

$$A \cdot S2 \text{ row1: } \begin{bmatrix} 0, 1 \end{bmatrix} \times \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1, 1 \end{bmatrix}$$

$$+ E2 \text{ row1: } \begin{bmatrix} 0, 0 \end{bmatrix} = \begin{bmatrix} 0, 0 \end{bmatrix} + \begin{bmatrix} 1, 1 \end{bmatrix} = \begin{bmatrix} 1, 1 \end{bmatrix}$$

$$A \cdot S2 \text{ row2: } \begin{bmatrix} 1, 1 \end{bmatrix} \times \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 2, 1 \end{bmatrix} = \begin{bmatrix} 0, 1 \end{bmatrix}$$

$$+ E2 \text{ row2: } \begin{bmatrix} 0, 1 \end{bmatrix} = \begin{bmatrix} 0, 1 \end{bmatrix} + \begin{bmatrix} 1, 1 \end{bmatrix} = \begin{bmatrix} 1, 0 \end{bmatrix}$$

$$A \cdot S2 \text{ row3: } \begin{bmatrix} 0, 0 \end{bmatrix} \times \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 0, 0 \end{bmatrix}$$

$$+ E2 \text{ row3: } \begin{bmatrix} 1, 0 \end{bmatrix} = \begin{bmatrix} 0, 0 \end{bmatrix} + \begin{bmatrix} 1, 0 \end{bmatrix} = \begin{bmatrix} 1, 0 \end{bmatrix}$$

$$A \cdot S2 \text{ row4: } \begin{bmatrix} 1, 0 \end{bmatrix} \times \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1, 0 \end{bmatrix}$$

$$+ E2 \text{ row4: } \begin{bmatrix} 0, 0 \end{bmatrix} = \begin{bmatrix} 0, 0 \end{bmatrix} + \begin{bmatrix} 1, 0 \end{bmatrix} = \begin{bmatrix} 1, 0 \end{bmatrix}$$

Step 2 Verify $b1 = A^T \cdot f1 \text{ mod } 2$

$$A^T: \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & 0 \end{bmatrix} \quad \begin{matrix} 2 \times 4 & 4 \times 1 \end{matrix}$$

$$f_1: \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$A^T \text{ row1: } (0 \times 1) + (1 \times 0) + (0 \times 0) + (1 \times 0) = 0$$

$$A^T \text{ row2: } (1 \times 1) + (1 \times 0) + (0 \times 0) + (0 \times 0) = 1$$

Step ④ Verify $b2 = A^T \cdot f2 \text{ mod } 2$

$$A^T: \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & 0 \end{bmatrix}$$

$$f_2: \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

$$A^T \text{ row1: } (0 \times 0) + (1 \times 0) + (0 \times 1) + (1 \times 1) = 1$$

$$A^T \text{ row2: } (1 \times 0) + (1 \times 0) + (0 \times 1) + (0 \times 0) = 0$$

Step ⑥ $k_1 = U_2^T \cdot f_1 + S_1^T \cdot b_2 \text{ mod } 2$

$$U_2^T: \begin{bmatrix} 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

$$f_1: \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad 2 \times 4 \times 4 \times 1$$

$$U_2^T \cdot f_1 = \begin{bmatrix} 1, 1 \end{bmatrix}$$

$$S_1^T: \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

$$b_2: \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$r_1: (0 \times 1) + (0 \times 0) = 0$$

$$r_2: (0 \times 1) + (1 \times 0) = 0$$

$$\begin{bmatrix} 1, 1 \end{bmatrix} + \begin{bmatrix} 0, 0 \end{bmatrix} = \begin{bmatrix} 1, 1 \end{bmatrix}$$

Step ⑦

$$k_2 = U_1^T \cdot f_2 + S_2^T \cdot b_1 \text{ mod } 2$$

$$U_1^T: \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

$$f_2: \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

$$r_1: (0 \times 0) + (1 \times 0) + (0 \times 0) + (0 \times 1) = 0$$

$$r_2: (0 \times 0) + (1 \times 0) + (0 \times 0) + (0 \times 1) = 0$$

$$S_2^T: \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

$$b_1: \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$r_1: (1 \times 0) + (1 \times 1) = 1$$

$$r_2: (0 \times 0) + (1 \times 1) = 1$$

Step ⑧

$$k_1 = \begin{bmatrix} 1, 1 \end{bmatrix} \quad (l=2)$$

$$k_2 = \begin{bmatrix} 1, 1 \end{bmatrix}$$

$\text{maj}(k_1)$

non-zero element in $k_1 = 2$

$$2 \geq 1 \rightarrow \text{maj}(k_1) = 1$$

$\text{maj}(k_2)$:

non-zero element in $k_2 = 2$

$$2 \geq 1 \rightarrow \text{maj}(k_2) = 1$$

$$\text{maj}(k_1) = \text{maj}(k_2) \rightarrow 1 == 1$$

Failed case ex:

$$k_1 = \begin{bmatrix} 1, 0, 1 \end{bmatrix} \quad (l=3)$$

$$k_2 = \begin{bmatrix} 0, 1, 0 \end{bmatrix}$$

$\text{maj}(k_1)$:

non-zero element = 2, $\frac{1}{2} = 1.5$

$$2 \geq 1.5 \rightarrow \text{maj}(k_1) = 1$$

$\text{maj}(k_2)$:

non-zero element = 1, $\frac{1}{2} = 1.5$

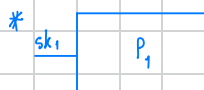
$$1 \geq 1.5 \rightarrow \text{maj}(k_2) = 0$$

1. Background

* Why we need the Key exchange

- symmetric key encryption
- Key distribution problem

* We need LWE Exchange to prevent middle attack person that he won't key from symmetric key



2. Current Challenges

* We already Key-Exchange is almost all communication - device !

* Most of them are not quantum-secure !

↓
if there is an attacker using quantum computer
the securities most of key exchange in our device will be broken

What we need is

3. Post-quantum Key Exchange

- Key exchange based on problem which is for quantum computers
- In this learning party with noise problem (LPN problem)
- LPN is to be hard in problem

4 LPN Problem

given: A : full-round binary $m \times n$ matrix

U = a binary $m \times l$ matrix

k =

output: S, E such that

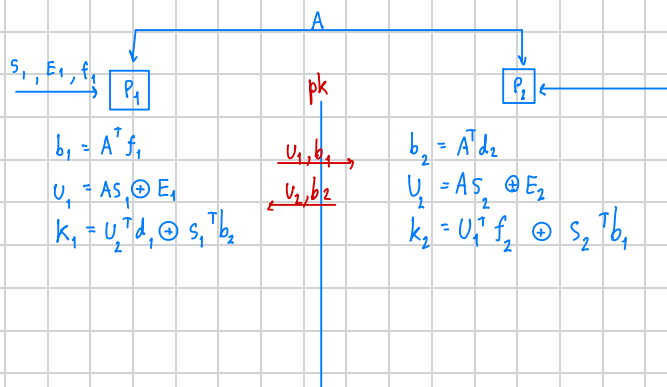
$$U = AS \oplus E$$

where S is a binary $n \times l$ matrix

E is a binary $m \times l$ matrix

such that each column has k non-zero elements

5 LPN based KE



6. Problems with LPN based ICE

★ Probability

$$k_1 = k_2$$

is too small !! (in 10,000 experiment - 0% !!!)

exp.	k_1	k_2
1		
⋮		
10,000		

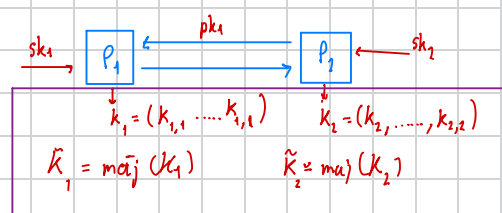
7. Our Solutions

- We refine K_1 and K_2 by using majority function

$$X = (x_1, \dots, x_n)$$

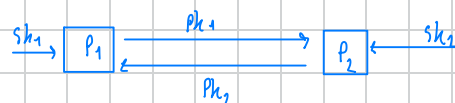
$$\text{maj}(X) = \begin{cases} 1 & \text{if the number of non zero } n \geq \frac{1}{2} \\ 0 & \text{otherwise} \end{cases}$$

8. 1st solution



Probability that $K_1 = K_2$ is $\approx \dots\%$ $\left(\begin{matrix} m = \\ n = \\ k = \\ l = \end{matrix} \right)$

9. 2nd solution majority of majority



$$\begin{aligned} K_1 &= (\text{maj}(k_1^{(1)}), \dots, \text{maj}(k_1^{(t)})) \\ K_2 &= (\text{maj}(k_2^{(1)}), \dots, \text{maj}(k_2^{(t)})) \\ \hat{K}_1 &= \text{maj}(K_1) \\ \hat{K}_2 &= \text{maj}(K_2) \end{aligned}$$

Prob that $\hat{K}_1 = \hat{K}_2$ is $\dots\%$

$$\left(\begin{matrix} m = \\ n = \\ k = \\ l = \\ t = \end{matrix} \right)$$

10. Conclusion

- We have implemented a post-quantum key exchange
- We have implemented that is KE exchange can run successfully with high probability using a certain parameter

11. Future direction

- the KE exchange is quite slow

1 mins $\approx$ 7 secs

$\cdot \uparrow \approx 10,000 \approx 2$ hrs

(CPU)
(CPU)
memory,

in 2 hrs we only agree on
1 bit !!

- According to the theory

$n = 256, m = 512$ is too small to guarantee the security

In, ... practice we need higher $n = m$

We need to improve the efficiency in algorithm

1 task

Single majority in same parameter with the 2nd task

↓
procedure
↓
result

2 task

→ slide
.. Nested major
Prob $k_1 = k_2 = \dots$
parameter

④ before LPN is hard given A, U, K is hard to find S and E that satisfied $U = AS \oplus E$

⑤

Until now not know how to LPN with computer yet
solve